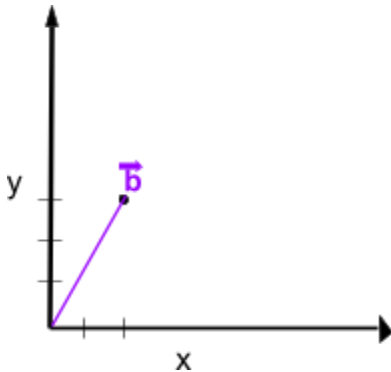


Vector Calculus

What is a vector?:

- The formal definition of a vector is an object that has both magnitude and direction.
 - They exist on a multi-dimensional coordinate system ie the cartesian coordinate system (x, y, z) but also can be expressed in polar coordinates (r,θ), etc.
 - Below is an example of a vector I labeled as $\vec{b}$.
- You can express $\vec{b}$ as $\vec{b} = 2\hat{x} + 3\hat{y}$ or $\vec{b} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $\vec{b} = (2, 3)$ or $\vec{b} = \langle 2, 3 \rangle$ or $\vec{b} = 2\hat{i} + 3\hat{j}$

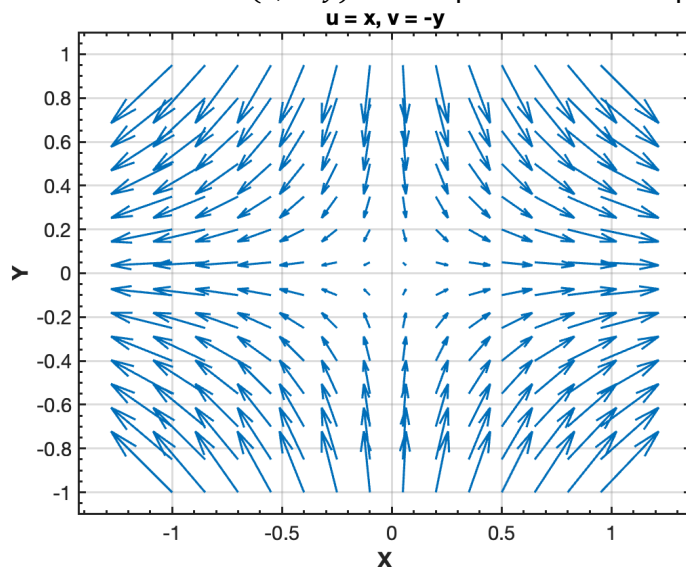


Vector Fields:

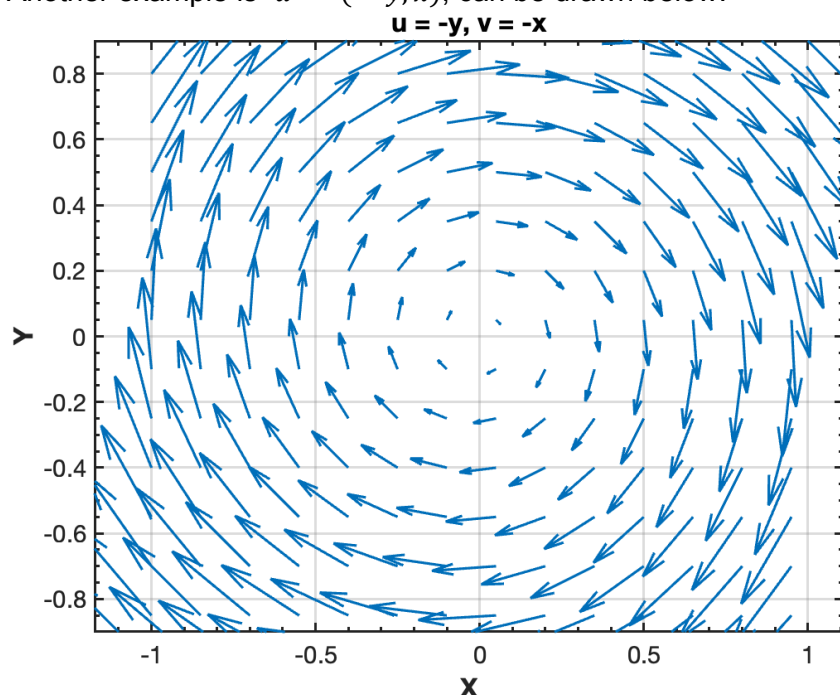
In oceanography, it is common to have velocity fields comprising vectors representing the speed (magnitude) and direction of fluid flow. Here $\vec{u}$ is commonly expressed with components $\vec{u} = (u, v, w)$.

[If new to vector fields, don't imagine a vector field as many small vectors, which can seem daunting and confusing, even though it is the best way to graphically represent them. A vector field is just a standard equation with a vector output instead of a scalar.]

A vector field $\vec{u} = (x, -y)$ can be plotted. An example is below:



Another example is $\vec{u} = (-y, x)$, can be drawn below:



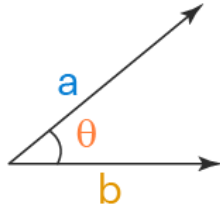
Vector Operations:

If $\vec{a} = (a_1, a_2, a_3)$ and $\vec{b} = (b_1, b_2, b_3)$.

- Vector Addition: $\vec{a} + \vec{b} = (a_1 + b_1, a_2 + b_2, a_3 + b_3)$
 - Two vectors can be added together, by adding each component to their respective dimensions, the output is another vector.
 - The same rules apply for subtraction.
- Scaling Vectors: $S\vec{a} = (Sa_1, Sa_2, Sa_3)$
 - The scalar is applied to each vector and the end result is a vector
- Magnitude of a Vector $\|\vec{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$
 - The result is a scalar that tells you the length of a vector
 - **Question: What is the magnitude of vector b? Answer: $\sqrt{13}$**

Dot Product:

- Two vectors can be dotted together to form a scalar
- $\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$
- $\vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\| \cos(\theta)$ where theta is the angle between vectors a and b

Dot Product

$$a \cdot b = |a| |b| \cos \theta$$

Cross-Product:

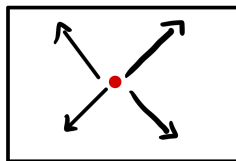
- Two vectors can be crossed together to form a vector orthogonal to the plane made of the two vectors

$$\begin{aligned}
 a \times b &= \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \\
 \Rightarrow a \times b &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} i - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} j + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} k \\
 \Rightarrow a \times b &= \langle a_2 b_3 - a_3 b_2 \rangle i - \langle a_1 b_3 - a_3 b_1 \rangle j + \langle a_1 b_2 - a_2 b_1 \rangle k
 \end{aligned}$$

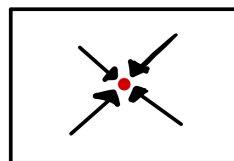
- Another formula is $\vec{a} \times \vec{b} = \|\vec{a}\| \|\vec{b}\| \sin(\theta) \hat{n}$
- Wait what does orthogonal mean?
 - When two objects are 90 apart
- Question: What is the dot product of 2 orthogonal vectors?
 - Answer: 0
- Question: At what angle do 2 vectors have the maximum cross-product?
 - Answer: $\pi/2$

∇ , The Gradient Operator - Now the Calculus comes in!!!

- $\nabla = \frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y} + \frac{\partial}{\partial z} \hat{z}$
- The gradient operator takes the partial derivative of each component in the corresponding dimension
 - **Gradient of a Function:**
 - $\nabla f = \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z}$
 - The direction of the gradient of a function denotes the direction of the greatest change of that function
 - f has to be a scalar function; ie only returns one value
 - Example:
 - $f(x, y) = x^2 + 3y + 1$
 - $\frac{\partial f}{\partial x} = 2x$ and $\frac{\partial f}{\partial y} = 3$ so:
 - $\nabla f = 2x \hat{x} + 3 \hat{y}$
 - **Question:** What is ∇f , if $f(x, y) = 2y \cos(\pi x)$ at $(x, y) = (4, 1/2)$
 - **Answer:** $\nabla f(4, 1/2) = (0, 2)$
 - **Divergence:**
 - Let $\vec{u} = (u, v, w)$
 - $\nabla \cdot \vec{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}$, this returns a scalar.
 - $\text{div}(\vec{u}) = \nabla \cdot \vec{u}$
 - What does divergence represent?
 - Think back to when we looked at vector fields. Divergence represents the collection or dispersion of fluid around a point
 - But, I think it is best understood when you draw it out.

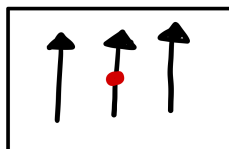


$$\nabla \cdot \vec{u} > 0$$



$$\nabla \cdot \vec{u} < 0$$

-
- **Question:** What is the divergence of this vector field around the point? **Answer:** 0

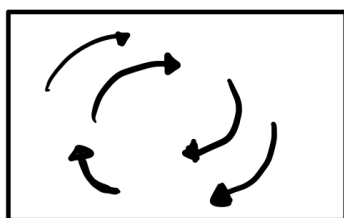


- **Divergence Continued:**

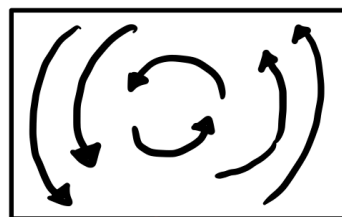
- When a fluid is incompressible (meaning constant density) the divergence is zero.

- **Curl/Vorticity:**

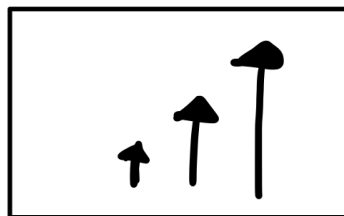
- Let $\vec{u} = (u, v, w)$
- $\nabla \times \vec{u} = \vec{w}$, $\vec{w} = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}, \frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}, \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$
- The curl at a point in a field is represented by a vector whose length and direction denote the magnitude and axis of the maximum circulation.
 - (think: where does it spin and how much does it spin)
- For a horizontal vector field:
 - Let $\vec{u}_h = (u(x, y), v(x, y))$
 - $\nabla \times \vec{u}_h = \vec{w}$, $\vec{w} = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \hat{z}$
 - You can use the right-hand-rule to discover the sign of the curl of your vector field, and axis of rotation
 - Below are some examples:



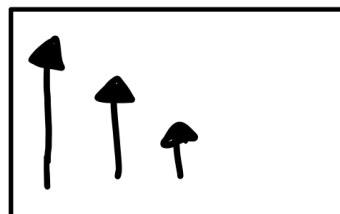
$$\nabla \times \vec{u} < 0$$



$$\nabla \times \vec{u} > 0$$

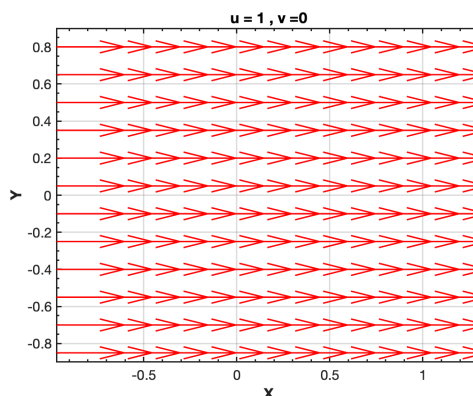


$$\nabla \times \vec{u} > 0$$



$$\nabla \times \vec{u} < 0$$

- Question: Draw a vector field with no curl
- Example Answer:



- **Laplacian Operator ∇^2 :**

- $\nabla^2 \vec{f} = \nabla \cdot \nabla \vec{f}$, This calculates the divergence of the gradient of $\vec{f}$ and returns a scalar
- $\nabla^2 \vec{f} = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$
- **Question: Calculate $\nabla^2 \vec{f}$ if $f = e^{kx} \sin(ky)$**
 - **Answer: 0**
- The Laplacian operator is seen in Laplace's equation which states $\nabla^2 \vec{f} = 0$. Solutions to Laplace's equation are called harmonic functions

Multi-Variable Integrals:

How to take a multi-variable integral:

- $\int_{y_1}^{y_2} \int_{x_1}^{x_2} f(x, y) dx dy = \int_{y_1}^{y_2} [\int_{x_1}^{x_2} f(x, y) dx] dy$
- Example: $f(x, y) = x^2 + 3y + 1$
 - $\int_0^4 \int_0^3 x^2 + 3y + 1 dx dy$
 - $\int_0^3 x^2 + 3y + 1 dx = \frac{27}{3} + 9y + 3$
 - $\int_0^4 9 + 9y + 3 dy = 9 * 4 + \frac{9}{2} * 4^2 + 3 * 4 = 120$

A way to imagine this integral is as a volume integral. Let's say we have a rectangular basin with a weird top. $f(x, y)$ is the variable height object and our integration bounds are the length and

width of the basin. The standard integral $\int_{y_1}^{y_2} \int_{x_1}^{x_2} f(x, y) dx dy$ as above then gives the volume of the object.

Surface Integral. (also called a flux integral)

-Say we have a 3d vector field and a surface (ie a flat plane) We want to find out how much our vectors align with the normal vector of the surface. (Remember the normal vector to a surface is the local orthogonal vector, and is a vector field existing on the surface)

-How can we do this?

-Oh wait! We learned about the dot product earlier, which is a measure of how much two vectors align with each other.

So, if we want to find the “total alignment” of a certain area of our surface, we should find an infinitesimal element of the *surface alignment* and integrate over it to get the total.

An area element $dA = dydx$. The direction we care about is the ‘normal.’ Call it $\hat{n}$. So an infinitesimal surface alignment element is $\vec{f} \cdot \hat{n} dA = \vec{f} \cdot \hat{n} dx dy$

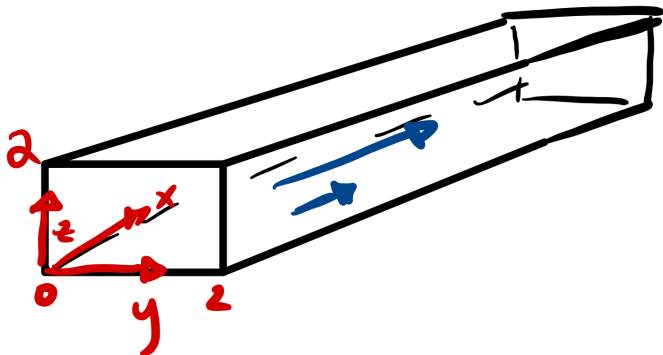
Integrate over the area bounds and we get the surface integral. This is also called the flux

$$Flux = \int_a^b \int_a^b \vec{f} \cdot \hat{n} dx dy$$

Flow through a Channel:

Flow $= \int \int \vec{u} \cdot d\vec{A} \hat{n}$, where S is the channel cross-section and A is the area normal to the plane of the flow field.

Example: $\vec{u} = (2z - 1, 0, 0)$ flow is in the x direction and varies by depth:



Here $d\vec{A} = dydz \hat{x}$ since the flow is in the x direction.

$$\text{Net Flow} = \int_0^2 \int_0^2 (2z - 1) \hat{x} \cdot \hat{x} dy dz$$

$$\text{Net Flow} = \int_0^2 \int_0^2 2z - 1 dy dz$$

First take the inside integral:

$$\int_0^2 2(2z - 1) dz = 2(2^2 - 2) = 4$$

Question: What could the units be?

Answer: m^3/s etc

Mass Calculation:

$$\iiint_V \rho(x, y, z) dV = \text{Mass if } \rho \text{ is density}$$

Divergence Theorem

$$\iiint_V (\nabla \cdot \mathbf{F}) dV = \oiint_S (\mathbf{F} \cdot \hat{\mathbf{n}}) dS.$$

The left side of the formula is an integration over a chosen volume. (Assuming we are working a three-dimensional problem). We are integrating over the divergence of our vector field $\mathbf{F}$.

A way to think about this is that we are going through every point in a chosen volume and finding the divergence of that point, and adding all of them up.

Now, a little hint about problems you might see is that we use this a lot for fluid like water, which is an incompressible fluid. Incompressible means that the divergence must be zero, as any vector lines pointed towards or away from each other would mean a density differential. So this left side is zero a lot.

The right side is just a standard surface integral as we learned earlier. It is still a measure of how much the vector field aligns with the normal of the surface we have chosen.

The important relation between the volume and surface integrals is that the surface integral must perfectly and completely surround the volume, or the volume must perfectly fill the surface, whichever way you want to put it. This means of course the surface must be closed.

Divergence Theorem Example:

What's the flux of $\vec{f} = (xyz, xyz, xyz)$ through a unit cube?

(Remember that the surface integral of the cube is the flux.)

$$\text{Flux} = \int_a^b \int_a^b \vec{f} \cdot \hat{\mathbf{n}} dx dy = \int_0^1 \int_0^1 (xyz, xyz, xyz) \cdot \vec{dA}$$

This is doable, but a total hassle. You have to do the flux through each face of the cube and add it up. Let's use the divergence theorem instead!

Flux=

$$\iiint_V \nabla \cdot \vec{F} dv = \int_0^1 \int_0^1 \int_0^1 (yz + xz + xy) dx dy dz = \int_0^1 \int_0^1 (yz + 1/2 z + 1/2 y) dy dz = \int_0^1 (z/2 + z/2 + 1/4) dz = 1/4 + 1/4 + 1/4 = 3/4$$